



U.S. Army Space and
Missile Defense Command

U.S. Army Space and
Missile Defense School

Operating in a Contested Space Environment

Cyber CoE

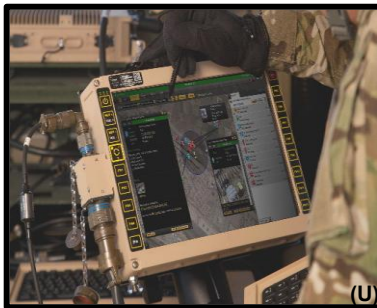
Army Space Training Division





Space Introduction

- **The Army is the largest user of space-enabled capabilities, integrated into all six Warfighting Functions**
 - Over 3200 pieces of BCT equipment are GPS enabled as compared to 500 pieces in a BN
 - Over 300 pieces of BCT equipment are SATCOM enabled as compared to 180 pieces in a BN
- **Satellites enable the Army Warfighter the ability to Maneuver and Attack with great precision (e.g., GPS), Communicate across the globe (e.g., SATCOM), see the battlefield (e.g., FFT), and Collect information throughout the AOR (e.g., ISR)**
- **Adversary capabilities will impact future Army engagements to operate in a Denied, Degraded, and Disrupted Space Operational Environment (D3SOE)**



FFT



SATCOM



GPS Guided Munitions



UAS providing ISR



LSA 1 - Identify Capabilities and Limitations of Space-Enabled Equipment

- **Positioning, Navigation, and Timing (PNT)**
- **Satellite Communications (SATCOM)**
- **Environmental Monitoring**
- **Intelligence, Surveillance, and Reconnaissance (ISR)**



Position Navigation and Timing (PNT)

GNSS NATIONS



GPS



GALILEO



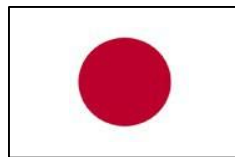
NAVIC



BEIDOU



GLONASS

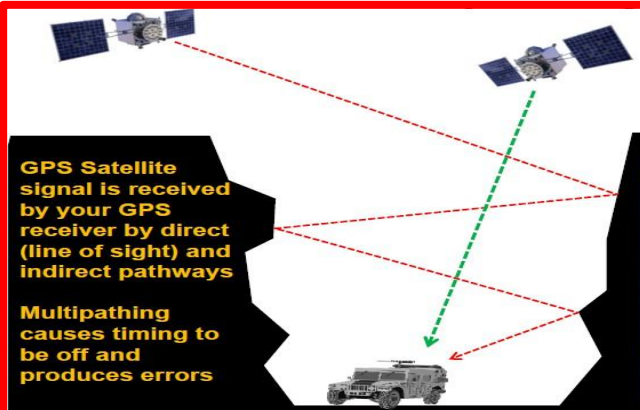


QZSS

- Each GPS receiver requires line-of-sight to a minimum of 4 satellites
 - Between 8-11 satellites in view at any given time
 - Latitude
 - Longitude
 - Altitude
 - Receiver clock bias (Time)
- GPS signal is an easy target
 - Very weak signal – ~ 50-watt light bulb
 - Frequencies used by GPS systems are widely known
 - GPS jammers are easy to acquire, build, and use
 - Most likely threat to GPS are ground-based jammers targeting receivers

THREATS TO NAVIGATION

Terrain Multit-Pathing



Navigation Jammers



Legacy Navigation



Satellite Communications (SATCOM)

SATCOM ADVANTAGES

Beyond Line-Of-Sight (BLOS)

voice / data / imagery / video

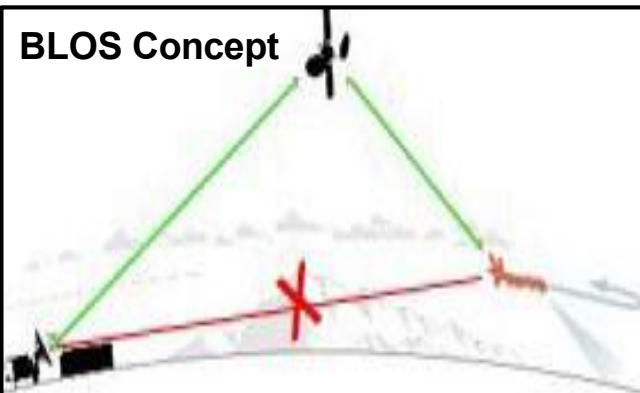
Extends Operational Range

strikes / ISR / mobility / re-supply

Personnel Recovery

Communications

BLOS Concept

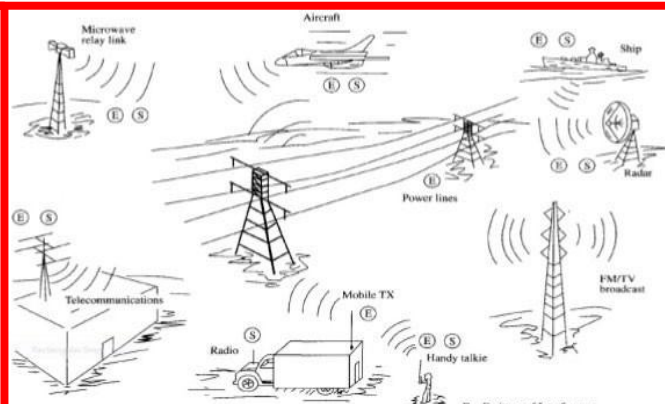
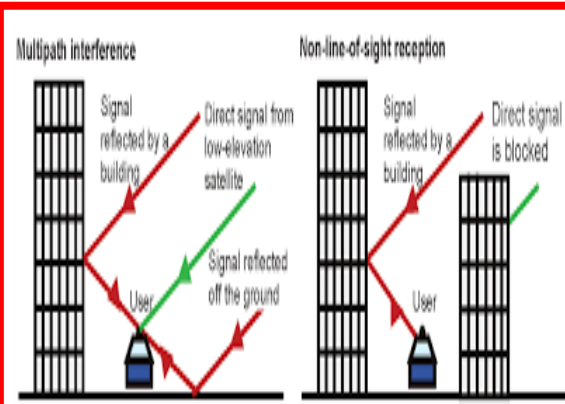


THREATS TO SATCOM

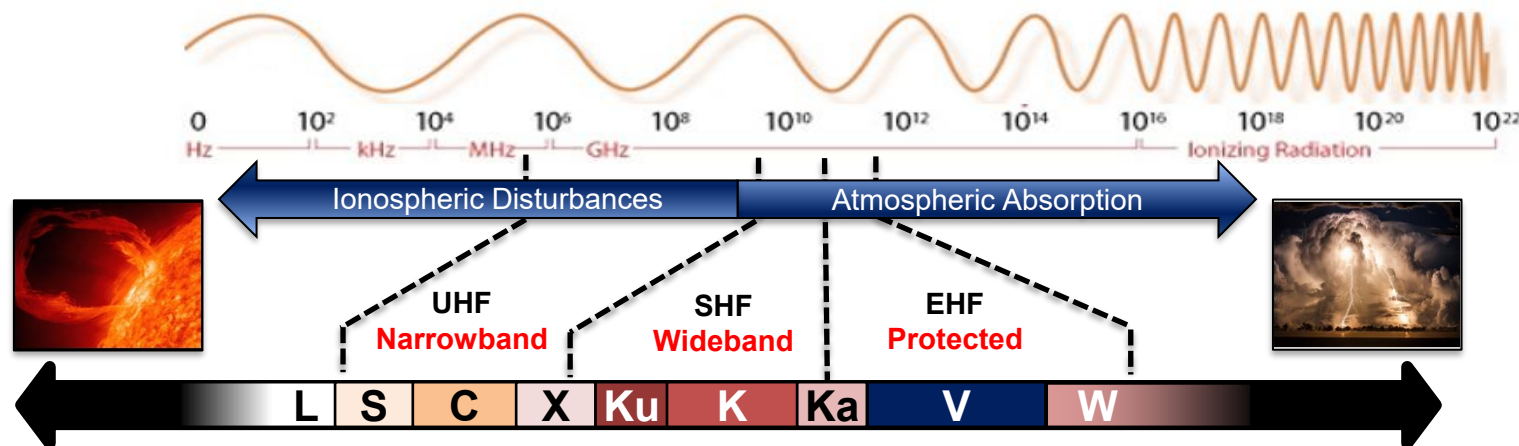
Urban/Mountain Terrain

Electromagnetic Interference

SATCOM Jammers



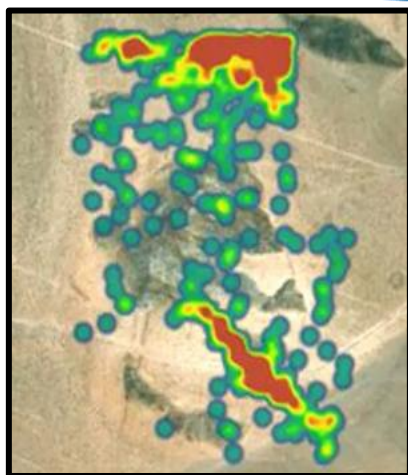
The Electromagnetic Spectrum



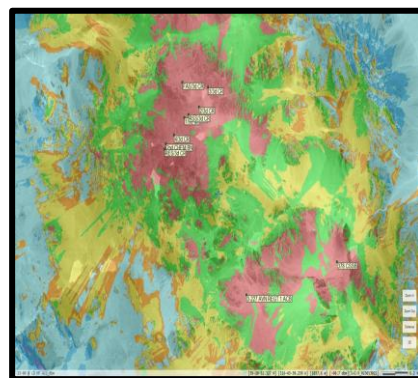
Accessibility

Anti-jam Capability

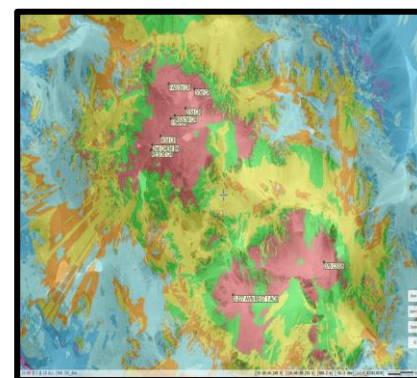
Bandwidth



Signal footprint of BN sized element



HF Comms Footprint 3-30 MHz



VHF Comms Footprint 30-300 MHz



Environmental Monitoring

Terrestrial Weather Satellites

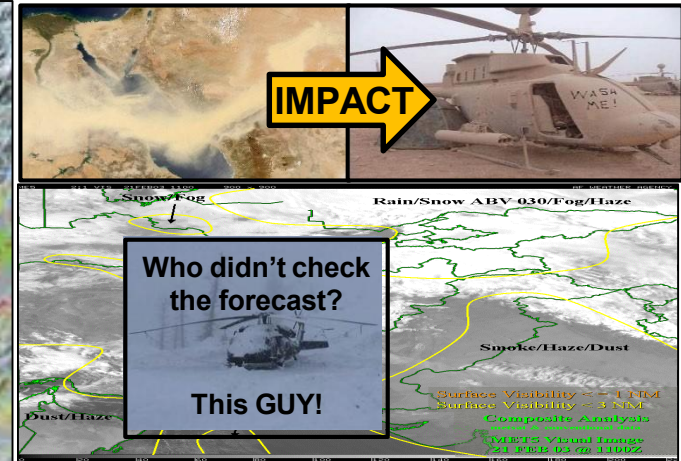
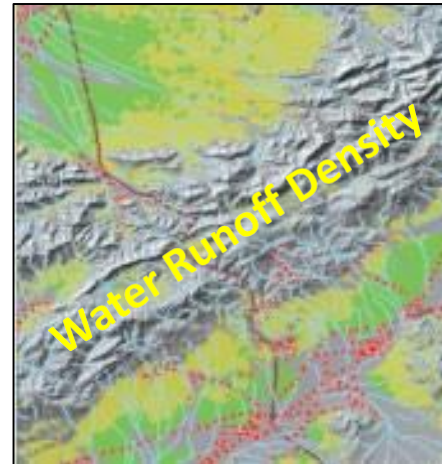
Easy public access



BLOS weather reporting Forecasts

Support mission planning

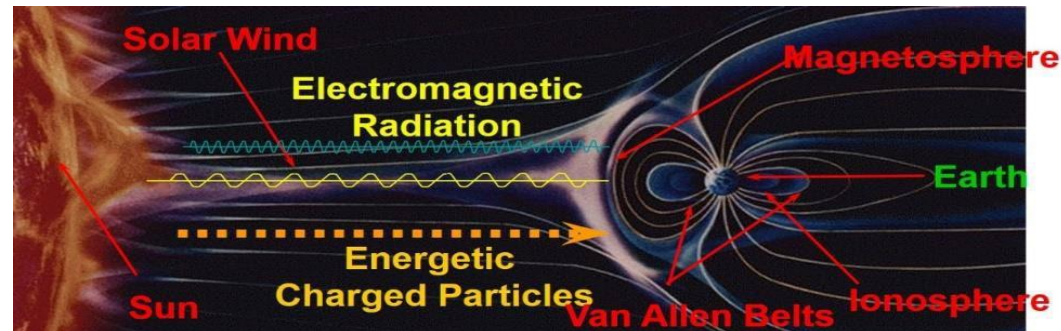
- LZ/DZ rain saturation / flood potential
- Sandstorm & Snowstorm Warnings



Space Weather Impacts

Disturbances in the space environment can degrade military systems

- Radar Systems
- Missile warning
- GPS Navigation (location errors)
- Intel Collection Systems
- HF Communications
- UHF/SHF SATCOM

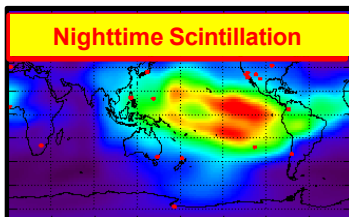
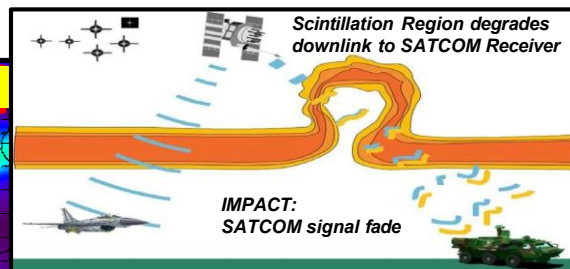


Aurora Borealis

Space Weather Phenomenon

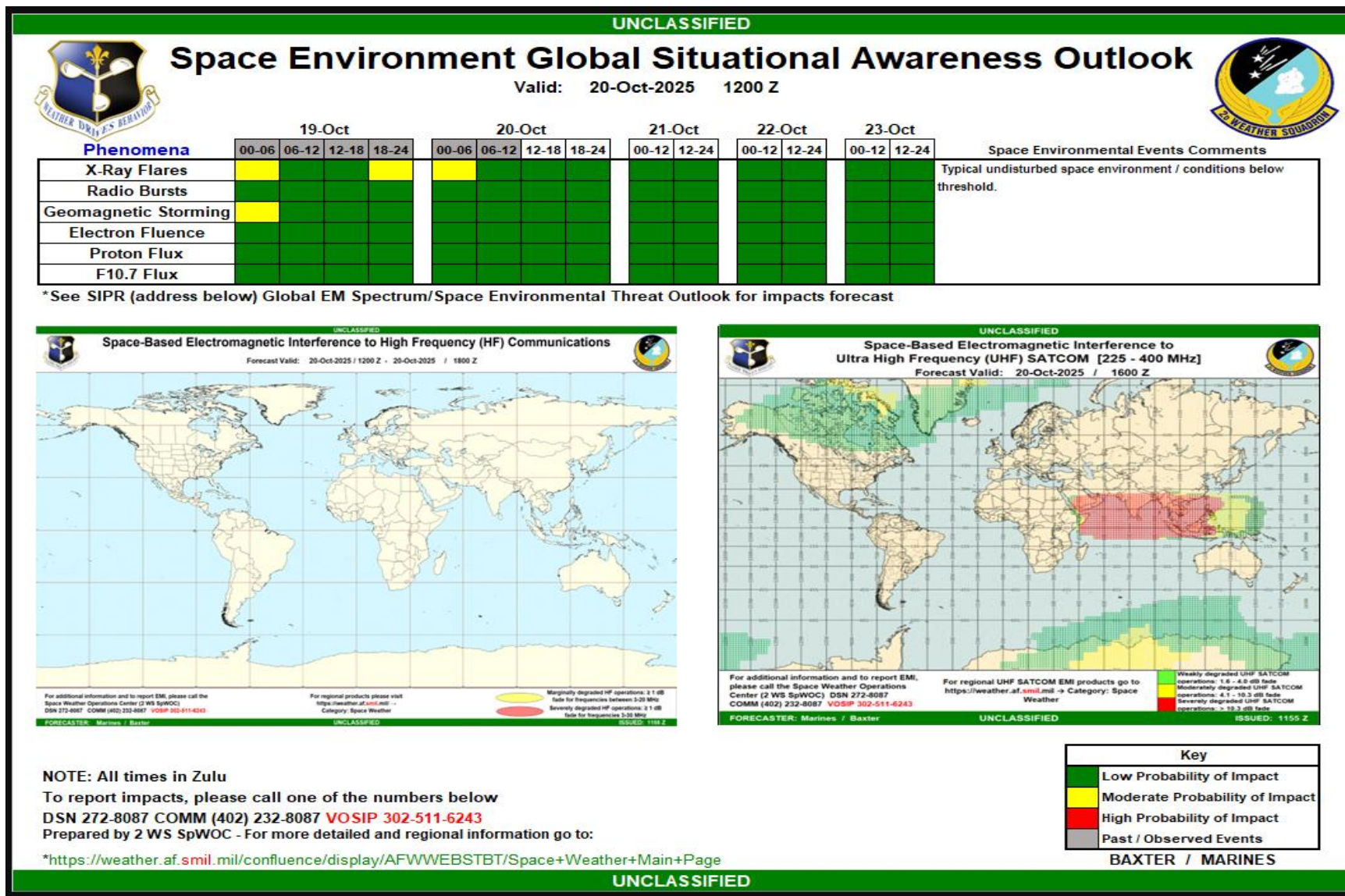
Electrically-charged electrons and protons collide with neutral atoms in the upper atmosphere

Colors indicate the atoms or molecules that are excited by the incoming particles





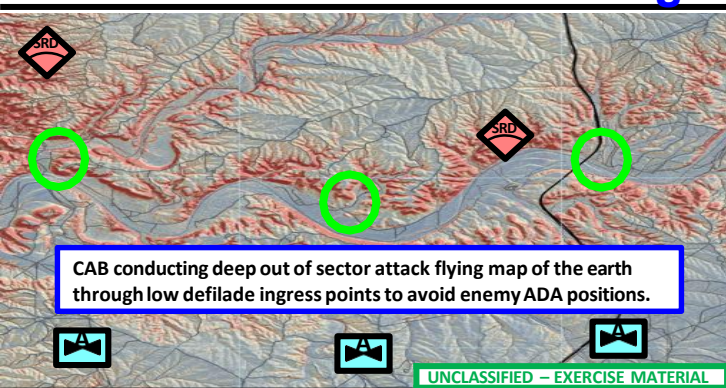
Example Space Weather Forecast



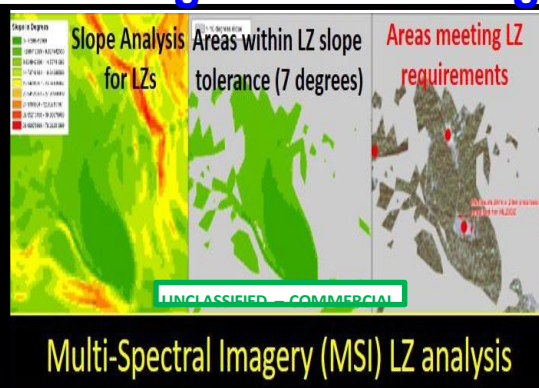
Satellite Imagery

GEOSPATIAL TERRAIN PLANNING

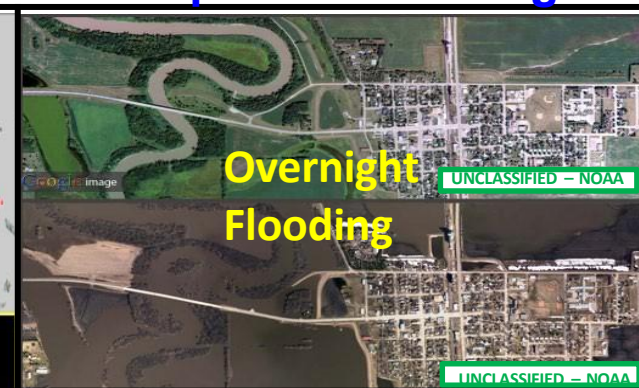
Maneuver/Obstacle Planning



Landing Zone Planning



Drop Zone Planning



THREAT IDENTIFICATION

Dispositions of Enemy Forces

Pan-Chromatic image

UNCLASSIFIED – COMMERCIAL

Multi-Spectral image

UNCLASSIFIED – COMMERCIAL

Detects camouflage nets

Enemy Infrastructure







Threats to Multidomain Operations

Adversary Threats

Lethal

Anti-Satellite Missiles

Attack on Space Ground Station

Non-Lethal

Electromagnetic Spectrum (EMS)

SATCOM Jammers ★

GPS Jammers ★

Electro-magnetic Pulse (EMP) Weapon

Cyberattacks Against U.S. Space Architectures

Lasers Against U.S. Satellites

Non-Adversary Factors

Non-Lethal

Communications Interference (Unintentional) ★

Space Debris

Solar Weather ★

Terrestrial Weather ★

Terrain and Vegetation ★

★ = Primary concern for Army tactical units

D3SOE is a CONDITION of the Operational Environment.

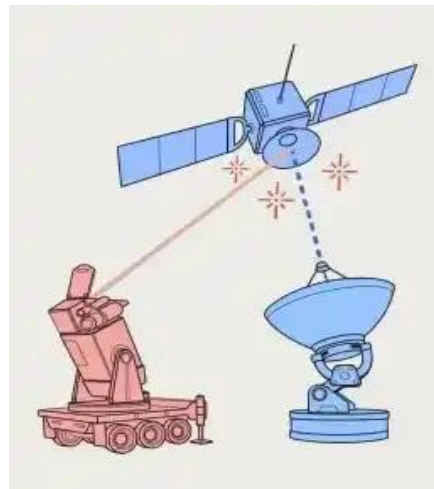


LSA 2 - Identify Space Linkages

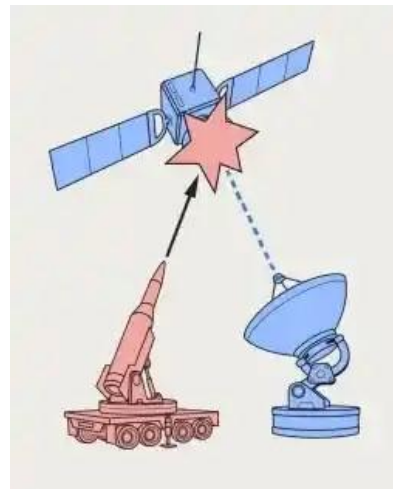
- **Adversary Threat Overview**
- **Space Impacts on Warfighting Functions**
- **Indicators and Mitigation**

Adversary Threat Overview

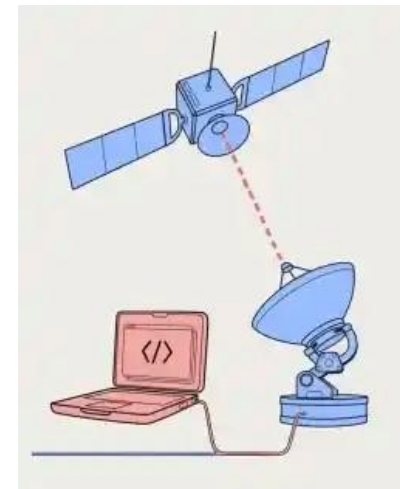
- **Peer Adversary**
 - Russia
 - China
- **Emerging Actors**
 - Iran
 - North Korea



Directed Energy Weapons

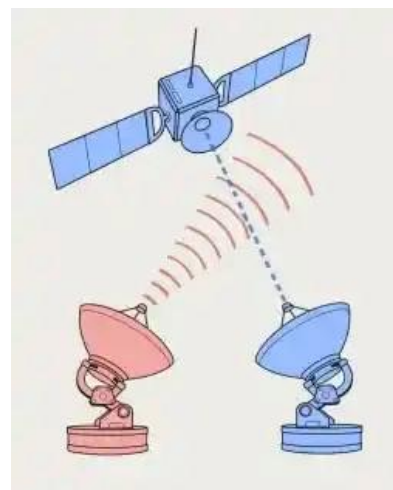


Direct-Ascent ASAT

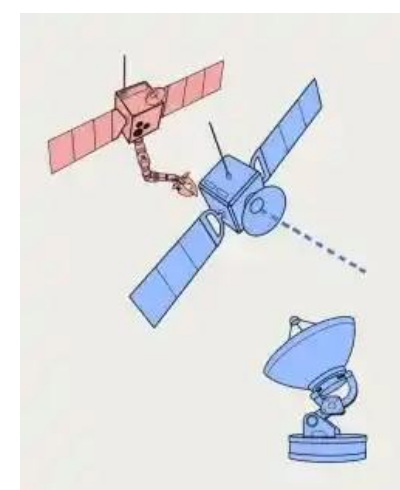


Cyber Attacks

- **Adversaries have been studying our dependence on SATCOM and GPS, possess ability to:**
 - Intercept and Disrupt Communications,
 - Locate Friendly Forces
 - Employ UAS for Collection and Targeting
- **GPS jammers are being actively used by adversaries** (inexpensive and worldwide availability)



Jamming Devices



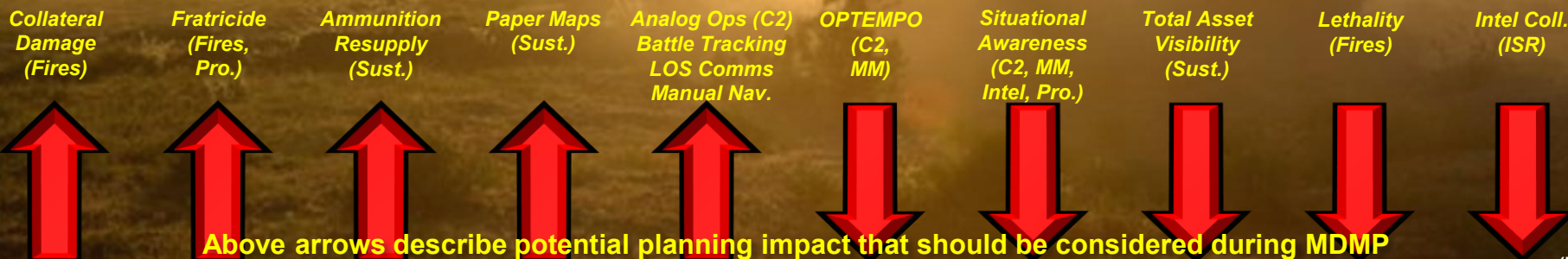
Co-Orbital ASAT



Space Impacts on Warfighting Functions

Space Capabilities	WARFIGHTING FUNCTIONS					
	<u>Movement and Maneuver</u>	<u>Fires</u>	<u>Intelligence</u>	<u>Command and Control</u>	<u>Protection</u>	<u>Sustainment</u>
<u>SATCOM</u>	Long distance unit synchronization	Effective and timely targeting	Reach back to higher level intel	Long distance C2	Timely warning	Pull/Push logistics
<u>PNT</u>	Accurate navigation - timely situation awareness	Precision targeting and tracking	Accurate ISR	Accurate COP	Reduces fratricide	Log tracking
<u>ISR</u>	Accurate adversary locations	Support to deep targeting	All source, and deep collection	Battlefield awareness	Early warning	Log planning
<u>Missile Warning</u>	Predicted impact point	Missile launch cueing	OPIR, Operational Environment Awareness	Operational Environment Awareness	TBM warning	Predicted impact point
<u>Weather, Terrain, & EM</u>	Operational planning, imagery for mobility	Operational planning	Weather	Weather	Operational planning	Operational planning
<u>Space Control</u>	Offensive, defensive	Offensive, defensive	Offensive, defensive, situational awareness	Offensive, defensive, situational awareness	Offensive, defensive	Offensive, defensive

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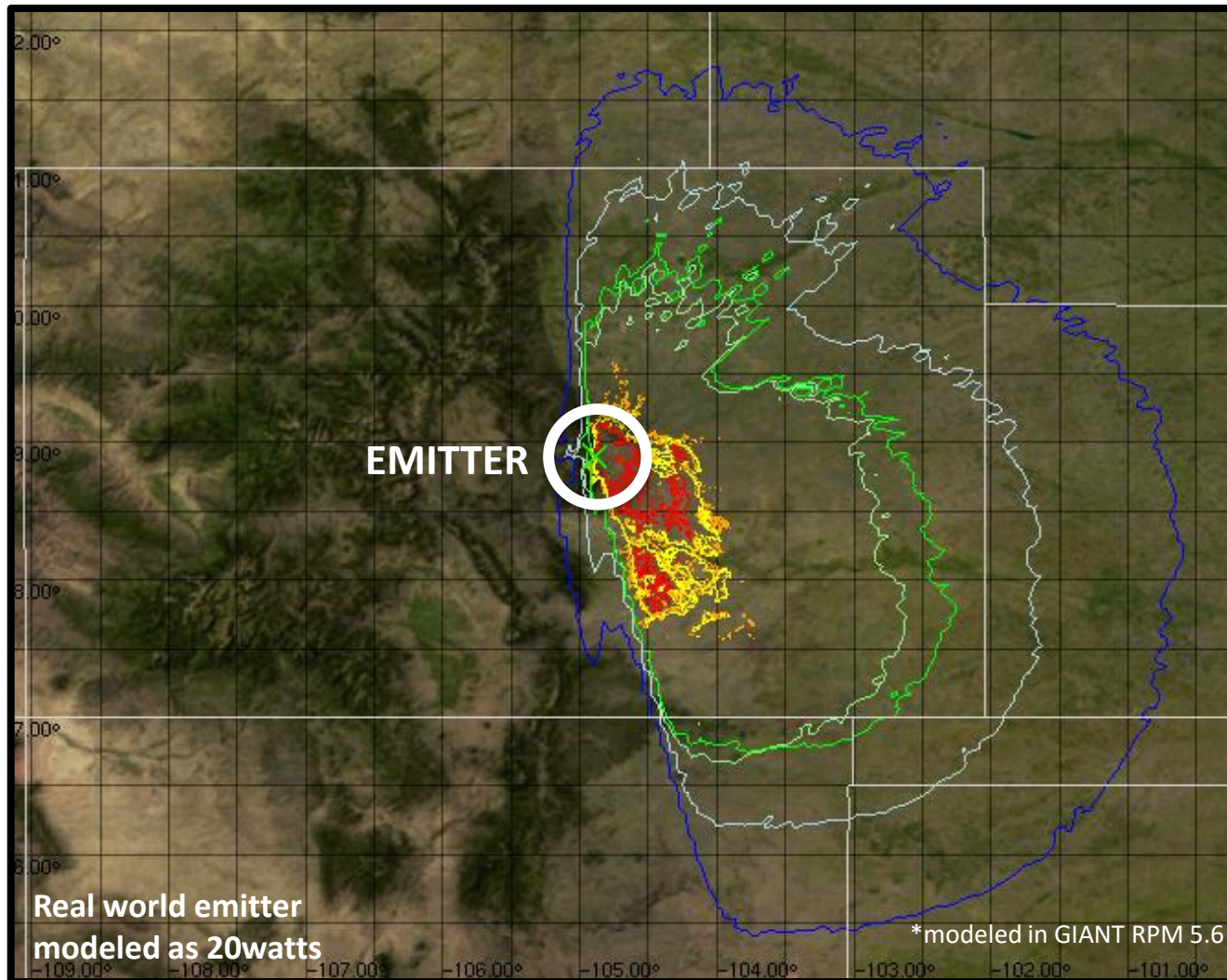


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Terrain Effects on Jamming

Overlay demonstrates the impacts of terrain on GPS jammers



UNCLASSIFIED



Contested Space Indicators

GPS Interference Indicators

- DAGR
 - “*Jamming Detected*” warning banner
 - Loss of 4 or more satellites displayed on DAGR
 - Increase in Jamming/ Signal ratio
- Friendly Force Tracking (FFT)
 - Display icon becomes stale
 - Location displayed does not match your map
 - “Gumball/ Stoplight” indicators change to red or amber



DAGR Jamming Detected Warning Banner



Body Shielding from Jammer

SATCOM Interference Indicators

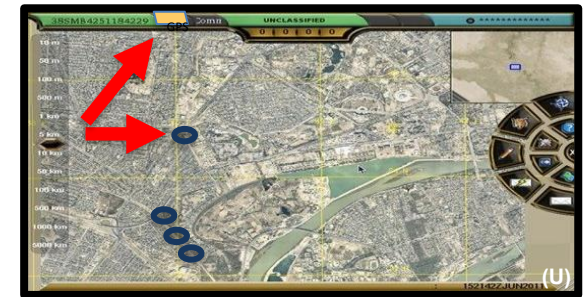
- Static (e.g., noise)
- Bleed over (e.g., voices/ “power hum”)
- Loss of communications (e.g., silence)

UAS Interference Indicators

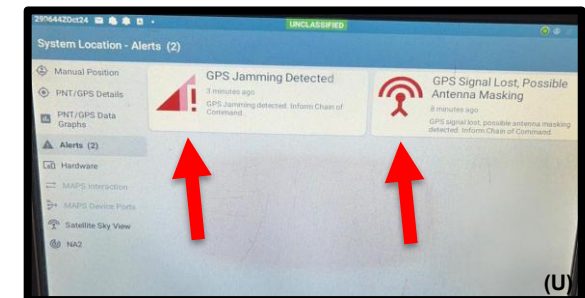
- Interference/ Jamming can cause:
 - UAS to can enter an orbit/ loiter
 - Delayed or frozen user interface causing crash
 - UAS to Return to Base (RTB), or revert to an undesired waypoint



**Aluminum can over
vehicle GPS
receiver - Good**



JCR/ JBC-P Display (Stale Icon)



New Software for JBC-P Being Contested



Mitigation Techniques for Contested, Congested, and Denied Environments



- **Emission Control (EMCON)**
 - Limit, discipline, and control electromagnetic emissions to reduce detection and targeting risk.
- **Redundancy & Layered Capabilities**
 - Employ multiple, diverse systems for communications, PNT, and ISR to ensure mission continuity.
- **Spectrum Management & Dynamic Spectrum Access**
 - Actively manage and adapt frequency use to reduce congestion and evade interference or jamming.
- **Hardening & Resilient Technologies**
 - Equip critical systems with anti-jam, secure, and hardened features to enhance survivability.
- **Deception & Signature Management**
 - Use decoys, false signals, and emission masking to mislead adversaries and protect critical assets.
- **Alternate Navigation & Timing Solutions**
 - Train and equip forces with methods for GPS-denied operations (e.g., inertial, RF, visual navigation).
- **Electronic Warfare (EW) Capabilities**
 - Employ offensive and defensive EW to disrupt adversary operations and protect friendly forces.
- **Training for Degraded & Denied Conditions**
 - Regularly exercise under space and EMS-denied scenarios to maintain readiness and adaptability.
- **Intelligence, Surveillance, and Threat Awareness**
 - Maintain persistent ISR and space situational awareness to detect and respond to emerging threats.



LSA 3 - Identify Space Threat Capabilities

- **D3SOE Lessons Learned**
- **MDMP Space Considerations**
- **Key Takeaways**



Lessons Learned in Ukraine

- **EW Integration in Response to Russia-Ukraine Conflict**

- Aggressive measures undertaken during Army's shift to Multidomain Operations and LSCO

- **Minimized Electromagnetic Emissions**

- Mitigation of risks through limiting detectable signals

- **Reduced Dependencies on Positioning, Navigation, and Timing (PNT) Technologies**

- Strengthening operational capabilities through diversification

- **Reintroduction of Tactical Ground EW Capabilities**

- Implementation of Terrestrial Layer System (TLS) as the primary EW and SIGINT weapon system

- **UAS (Drones)**

- The Ukrainian battlespace features the most intensive use of drones in a military conflict in history, marking a shift in warfare tactics and technology. They are equipped with different types of explosives and have destroyed hundreds of Russian tanks, missile systems, and other vehicles.

- **Starlink**

- Starlink became critical to the defense effort by Ukrainian military by having access to satellites and ultimately the portability, cost, and ability to resist jamming. This includes the fact that it is easy for an adversary to shut down a single satellite, but proliferated systems such as SpaceX's thousands of Starlink internet satellites, are much more resilient in the face of attacks.

- **Geolocation (Cell Phones and Social Media)**

- Always consider operations security when engaging on social media and use of cell phones in combat. Russia has unmanned aircraft dedicated to emulating cell phone towers to exploit this vulnerability. Russians are using electromagnetic warfare systems to locate Ukrainian forces with systems like the Leer 3 that can pick up more than 2,000 phones within a 3.7-mile range.



Starlink Used in Ukraine



FPV Drone (U)



Space Considerations During MDMP

The Military Decision-Making Process in a Denied, Degraded, and Disrupted Space Operational Environment

Annex N - Space Operations

Space Planning Considerations

- Who is the threat?
- Are SATCOM and PNT capabilities optimally postured to support operational mission accomplishments and integration into WfF?
- Will ISR satellites be available and provide timely and relevant data to support route planning?
- What are the terrestrial and space weather forecast, will there be impacts on operations during the planning and operation periods?

Step 1: Receipt of Mission
Identify Space as a Key Factor

Step 2: Mission Analysis
Space Threats & Risk Mitigation

Step 3: COA Development
Integrating Space Resilience

Step 4: COA Analysis (Wargame)
Simulating Space-Denied Scenarios

Step 5: COA Comparison
Space Resiliency as a Decisive Factor

Step 6: COA Approval
Command Guidance on Space Contingencies

Step 7: Orders Production
Integrating Space into OPORDs



Takeaways

PREPARE > RECOGNIZE > REACT > REPORT

- P** {
 - No Civilian GPS Receivers
 - Encrypt! Encrypt! Encrypt!
 - Know Your Equipment (uncontested vs contested)
 - Do Terrain Analysis in Planning
 - Have a PACE Plan
 - Brief Threats to the Lowest Levels
- R** {
 - Pay Attention to System/Device Indicators
- R** {
 - Use DAGR Jammer Finder App
 - Employ Mitigation Techniques- Body Shielding, Dig a Hole, Terrain
 - Move Out of the Denied Environment
 - Employ Direction Finding Techniques When Possible
- R** {
 - Report EMI & Enemy Electronic Contact to S2, S6, and EW



OPERATIONS

What We Do

The Operations Cell synchronizes the training, administrative, and operational efforts of the four branches. The team manages observations, data collection, lessons learned, and overall knowledge management for the Division. This cell assures that training teams and instructors have the training support resources available and requested for the various missions in support of the Total Army.

Specific Activities:

- ♦ Manages contested space effects processes, procedures, and validation
 - ♦ Global Positioning System
 - ♦ Multi-Global Navigation Satellite Systems
 - ♦ Joint Battlefield Command-Platform (JBC-P) & Joint Capabilities Release (JCR)
 - ♦ Wideband Authorized Injection Signal (Satellite Communications)
- ♦ Interfaces and collaborates with key organizations to assure opportunity and consistency of Army training and readiness
 - ♦ USSTRATCOM
 - ♦ Army Frequency Management Office
 - ♦ Army Spectrum Management
 - ♦ Army Space Support Elements
 - ♦ US Space Force
 - ♦ Department of Transportation
 - ♦ Federal Aviation Administration
- ♦ ASTD-internal Lessons Learned Management
- ♦ Management and Endorsement of Personnel Development Skill Identifier (PDSI) - S1A (Unit Space Trainer)
- ♦ Cross-Organizational Collaboration within the Space and Missile Defense School (SMDS)

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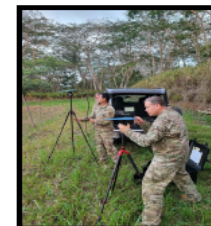
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Center Of Excellence

SPACE AND MISSILE DEFENSE SCHOOL



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[HTTPS://ATN.ARMY.MIL/SPACE-TRAINING-INTEGRATION-D3SOE/ARMY-SPACE-TRAINING-DIVISION](https://atn.army.mil/space-training-integration-d3soe/army-space-training-division)



HOME STATION TRAINING (HST)

What We Do

The HST Branch provides space training across the Total Army focused on a D3SOE and space enabling capabilities. Training packages are flexible and tailorable to unit requirements and constraints to maximize training opportunities and better prepare units deploying to a MCTC rotation or in support of combat and contingency operations abroad.

Available Training Options

I. Classroom Instruction:

- ◆ Space Basics & Use of Capabilities
- ◆ SATCOM & Electromagnetic Spectrum (EMS)
- ◆ Positioning, Navigation, & Timing (PNT)
- ◆ Intelligence, Surveillance, & Reconnaissance
- ◆ Planning Considerations & Jamming Mitigation
- ◆ Classified Threat Brief

II. **KLE/LPD.** Training provided to BDE and BN Senior Leaders and Staffs focusing on space domain considerations, and training expectations.

III. **Staff Planning.** Classroom instruction covering threats and space considerations from Staff and Warfighting Functions perspectives.

IV. **Live Effects Range.** Warfighters are afforded the opportunity to observe PNT denial effects and indicators on PNT-enabled equipment and put mitigation techniques into practice. ***Requires unit to request with a minimum of 90 days of advanced notice and coordination.**

V. **FTX Integration.** Integration into a unit's FTX training scenarios by providing live counter-space effects. Trainers observe, coach, and train Leaders, Staffs, and Warfighters under contested space domain conditions. ***Requires unit to request with a minimum of 90 days of advanced notice and coordination.**

*Training support funding is mostly internal to ASTD. Based on mission growth and limited resources, ASTD has a training priority list. Please inquire with HST Branch POC.

COMBAT TRAINING CENTERS (CTC)

Maneuver Combat Training Centers (MCTC)

What We Do

The MCTC Branch deploys to all MCTC rotations to support D3SOE exercise scenarios. Teams provide support at the National Training Center (NTC), Joint Readiness Training Center (JRTC), Joint Multinational Readiness Center (JMRC), and Joint Pacific Multinational Readiness Center (JPMRC). Support is also provided at National Guard Training Centers upon request.

The MCTC Branch is fully integrated into many training scenarios, exposing rotational training units to a D3SOE through the use of multiple GPS denial training devices. Teams Observe, Coach/Train (OC/T) leaders, staffs, and Soldiers to prepare for, recognize, react, and report in a D3SOE. They also provide subject matter expertise for Army units to understand, plan for, and capitalize on space-based capabilities.

Mission Command Training Program (MCTP)

What We Do

The MCTP Branch provides training support to Corps, Division, and Army Service Component Command (ASCC) Commanders and staffs. This training includes support to OPLAN/OPORD development, Mission Command Training, and OC/T support during Warfighter exercises. When requested and able, the team provides tailorable training during MCTP pre-exercise academic sessions.

- ◆ US Army Reliance on Space
- ◆ D3SOE
- ◆ Army Space Training Strategy
- ◆ US & Threat Counter-Space Capabilities
- ◆ D3SOE CTC Trends
- ◆ D3SOE Staff Planning, Execution, Considerations, & Tasks by Warfighting

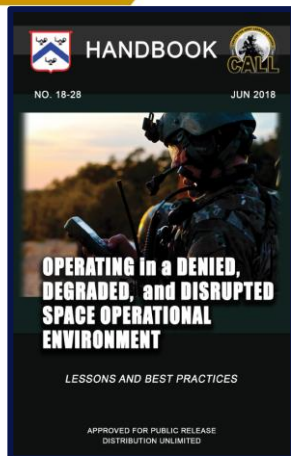
FOUNDATIONAL SPACE EDUCATION

What We Do

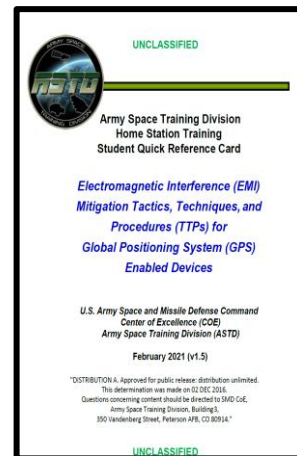
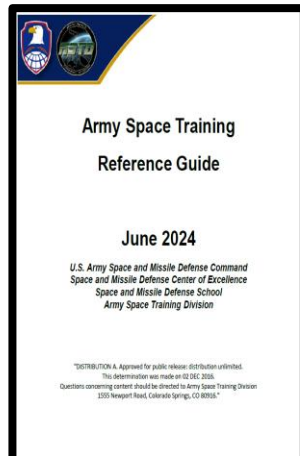
The Foundational Space Education Branch (FSEB) integrates foundational and contemporary space content to TRADOC CoEs and the Combined Arms Center enabling future integration at Initial Military Training (IMT) and Professional Military Education (PME). The FSEB provides subject matter experts to ensure Soldiers and military members, at all levels, are educated to operate effectively in a contested and congested Electromagnetic Spectrum (EMS) environment during Multidomain Operations. Simultaneously, the FSEB will ensure all trained Soldiers have access to all training material, space related products, and space points of contact for future integration at their home station. If provided the opportunity, the FSEB will assist in space content integration with Joint Forces.

Objectives

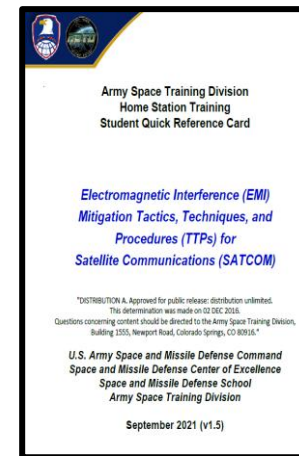
- ◆ Maintain close coordination with Army University/Combined Arms Center (CAC) for the purpose of Space/D3SOE integration into Captains Career Course (CCC), Basic Officer Leaders Course (BOLC), Warrant Officer Basic Course (WOBC), and Noncommissioned Officer Academy (NCOA) common core.
- ◆ Conduct Train-the-Trainer for resident CoE instructor cadre to effectively instruct space lessons.
- ◆ Provide Branch/CoE specific recommendations for D3SOE integration into technical phases of instruction.
- ◆ Provide space reference linkages for use by resident CoE instructor cadre
- ◆ Provide tailored/technical D3SOE guest instruction to select CoE to include:
 - ◆ Warrant College



**CALL D3SOE
Handbook**



EMI TTP GPS



EMI TTP SATCOM

Questions?

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